AVEECK PANDEY

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EDUCATION

SRM Institute of Science and Technology Chennai, India

Bachelor of Technology in Computer Science and Engineering *Jun 2022 – May 2026*

- CGPA: 8.36/10.0

EXPERIENCE

Rise Against Hunger Chennai, India

Software Development Intern *May 2024 – June 2024*

- Engineered an Inventory Management System with QR code scanning, reducing manual data entry time by **~60%** for over **500+ inventory items** tracked in real time
- Contributed to **3+ Student Resource Centre (SRC)** community initiatives, coordinating with a team of **10+ volunteers** to support organizational mission

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, JavaScript, SQL
Frontend Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Design
Backend Development: Node.js, Express.js, REST APIs, WebSockets
Databases: MySQL, MongoDB, PL/SQL, Weaviate (Vector Database)
AI/ML: TensorFlow, Scikit-learn, Transfer Learning, RAG Pipelines, Gemini API
Cloud & DevOps: AWS EC2, Docker, Git, Postman, VS Code, Streamlit

TECHNICAL PROJECTS

Breast Cancer Detection | Python, TensorFlow, Scikit-learn |_____ 2024

- Engineered a multi-model ML classification system detecting breast cancer tumors as malignant or benign, achieving **96% accuracy** on the Wisconsin Breast Cancer dataset (569 samples)
- Enabled users to select and compare **3+ ML models** (Logistic Regression, SVM, Random Forest) with real-time performance metrics for each, reducing evaluation time by **40%**
- Executed end-to-end data preprocessing and feature selection across **30 clinical features**, deploying an interactive Streamlit interface for real-time diagnostic predictions

Chatty — Real-time Chat Application | React.js, Node.js, WebSockets, MongoDB _____ 2024

- Architected a full-stack messaging application with real-time bidirectional communication using WebSockets and Socket.io, achieving message delivery latency under **100ms**
- Implemented user authentication, chat rooms, and persistent message history using MongoDB, supporting **50+ concurrent users** without performance degradation and achieving **99.9% uptime** in testing
- Designed a responsive UI with **5+ core features** (media sharing, user profiles, message history) delivering seamless UX across desktop and mobile browsers

Skin Disease Prediction System | Python, TensorFlow, MobileNetV2 _____ 2024

- Constructed a deep learning model using **MobileNetV2** (transfer learning) to classify **7 skin conditions** from dermoscopic images, achieving **92% accuracy**
- Applied data augmentation across a dataset of **10,000+ images** to improve model generalization across diverse skin types and reduce overfitting by **~18%**
- Streamlined an end-to-end prediction pipeline with image preprocessing and a clinical-grade web interface, cutting diagnosis turnaround time to under **2 seconds**

ML Research Assistant | Python, Weaviate, Gemini API, Streamlit, AWS EC2 _____ 2024

- Designed a production-grade Retrieval Augmented Generation (RAG) pipeline indexing **10,000+ ML research paper chunks** using Google Gemini API as the core language model
- Integrated Weaviate Cloud as the vector database, enabling semantic retrieval with **sub-second query response times** across Hugging Face paper embeddings
- Deployed the application on **AWS EC2** with a Streamlit front-end, managing dependencies via UV and reducing environment setup time by **50%** over traditional pip workflows

CERTIFICATIONS

- NPTEL Cloud Computing Certificate – IIT Kharagpur, 2025 (Elite category)